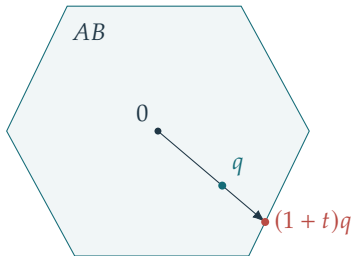


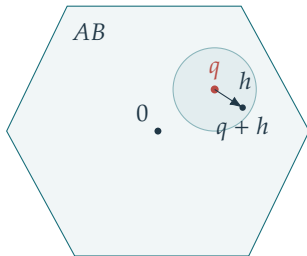
Radial extension from an interior point



$$\begin{aligned} Au &= (1+t)q, u \in B \\ z &= ru/(1+t) \text{ gives} \\ Az &= Ax_0 \text{ and } \|z\|_1 < r. \end{aligned}$$

$$r = \|x_0\|_1 > 0, \quad q = Ax_0/r, \quad A \text{ onto}$$

A better feasible vector gives an interior point



$$\begin{aligned} Az &= Ax_0, \|z\|_1 < r \\ q + h &= A(z/r + A^+h) \in AB \\ &\text{for small } h. \end{aligned}$$